COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 007	**SECTION:**	CPE11S4
DATE PERFORMED:	08/7/2026	**DATE SUBMITTED:**	08/7/2026
NAME:	Mangana, Prince Adrian M.	**INSTRUCTOR:**	Robin Valenzuela

ACTIVITY NO. 3.1 Control Structures (Part 2)

1. PROCEDURE OUTPUT

Perform each of these steps for the below problems:
1. Try to create an output of the following:
a. for (i = 1; i <= 10; i++)

for.cpp > main()

```cpp
#include <iostream>
using namespace std;
int main (){


    for (int i = 1; i <= 10; i++){
        cout << i << endl;
    }

    return 0;
}
```

```
1
2
3
4
5
6
7
8
9
10
PS C:\Users\admin pc\OneDrive\Documents\1styear.cpp>
```

b. for (i = 100; i>= 1; i--)

```cpp
#include <iostream>
using namespace std;
int main (){


    for (int i = 100; i>= 1; i--){
        cout << i << endl;
    }

    return 0;
}
```

```
PS C:\Users\admin p
100
99
98
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11
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5
4
3
2
1
PS C:\Users\admin pc\OneDrive\Doc
```

c. for (i = 7; i <=77; i += 7)

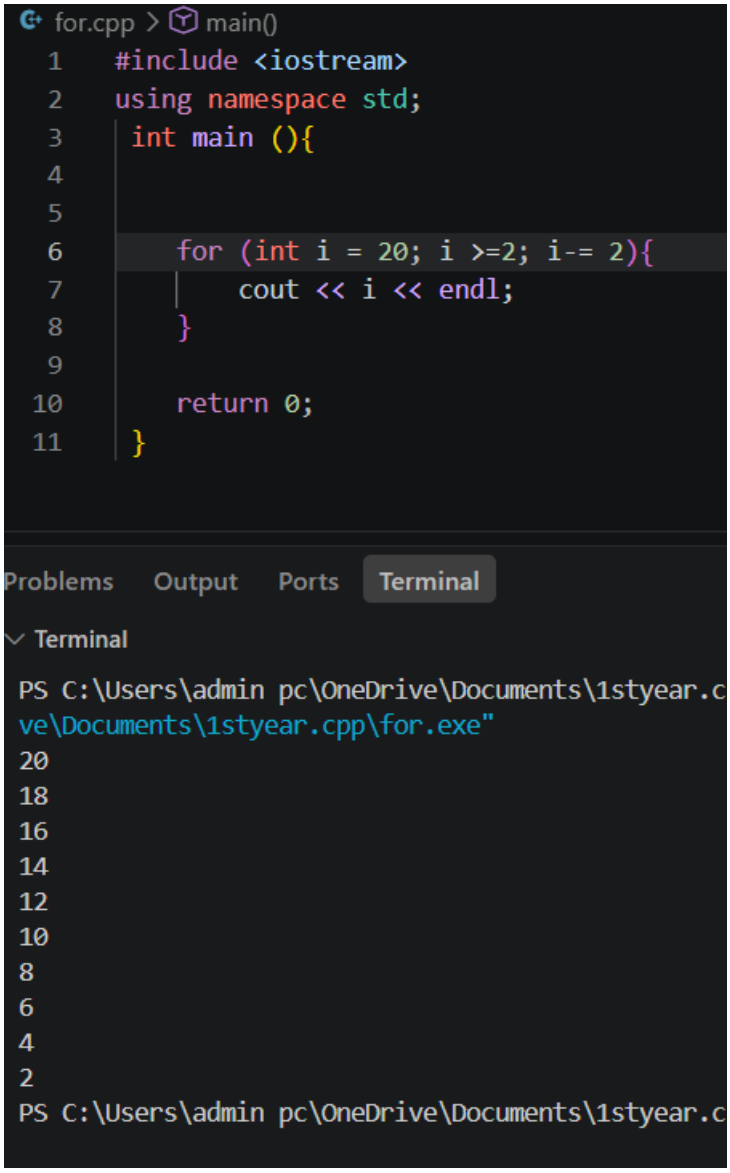
```cpp
#include <iostream>
using namespace std;
int main (){


    for (int i = 7; i <=77; i += 7){
        cout << i << endl;
    }

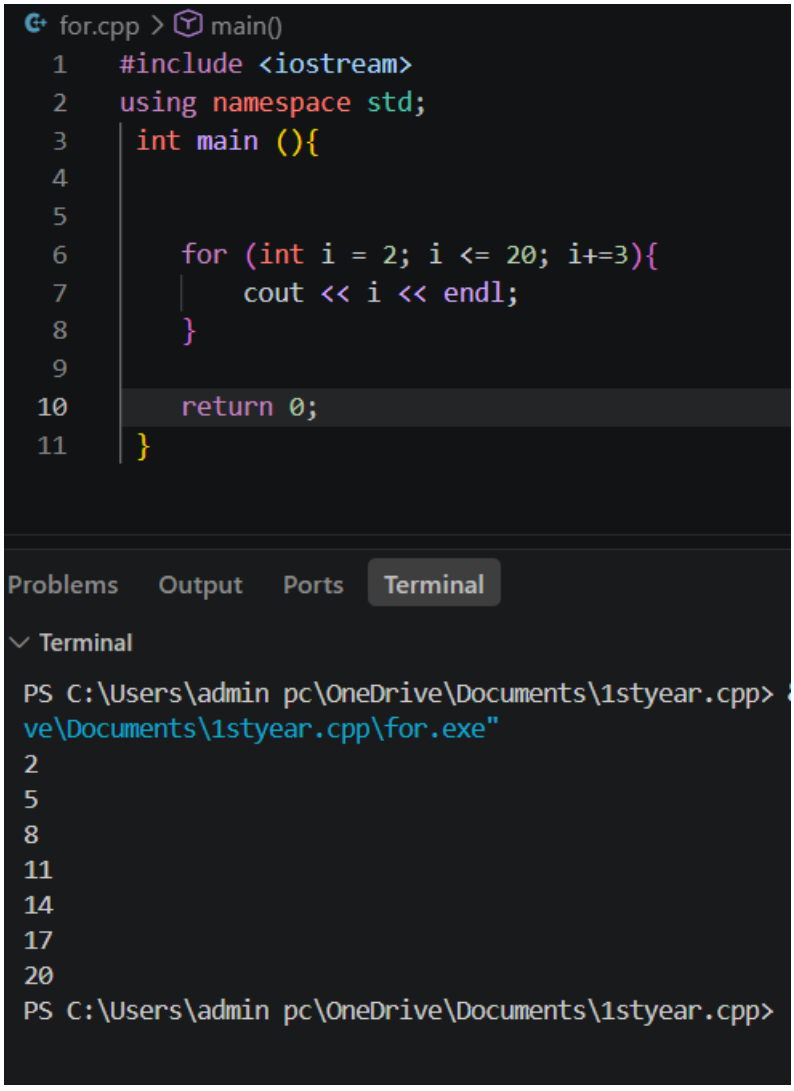
    return 0;
}
```

```
PS C:\Users\admin pc\OneD
ve\Documents\1styear.cpp\
7
14
21
28
35
42
49
56
63
70
77
PS C:\Users\admin pc\OneD
```

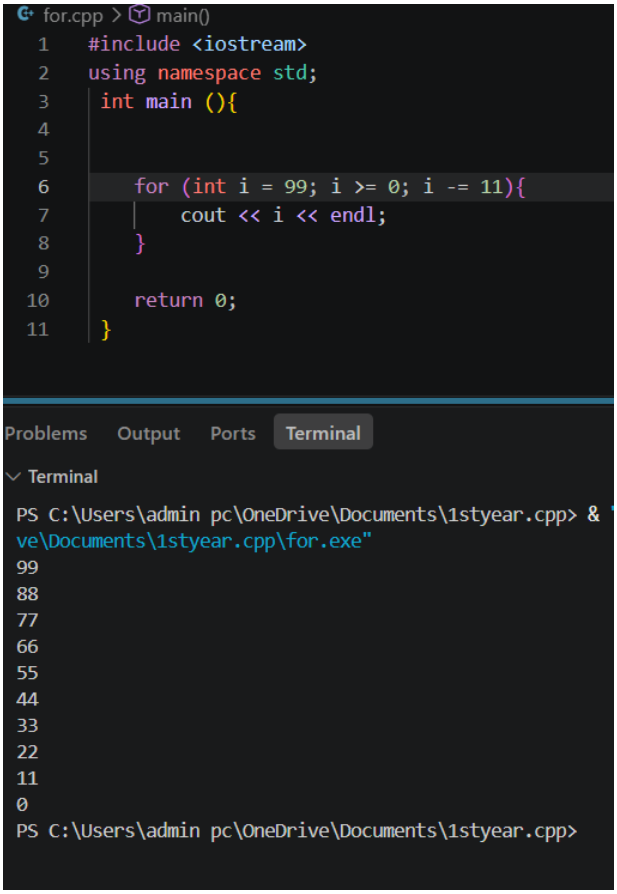
d. for (i = 20; i >=2; i-= 2)



e. for (i = 2; i <= 20; i+=3)



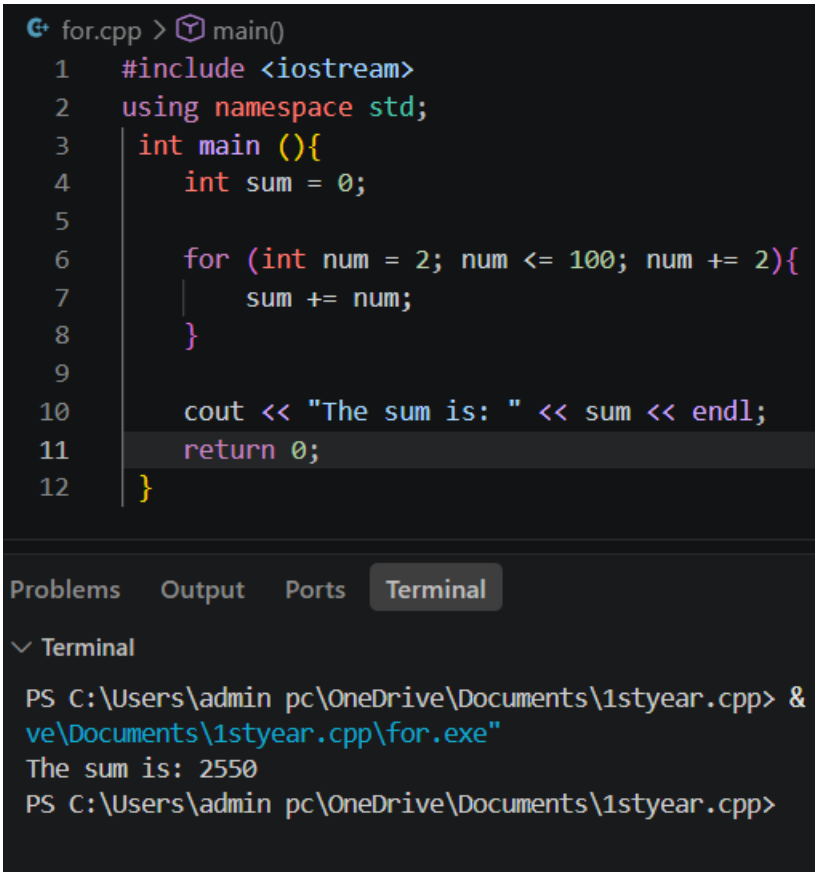
f. for (i = 99; i >= 0; i -= 11)



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

For each of the following, create the following c++ code:

1. Create a program that uses for structure to sum all the even integers from 2 to 100. Sum is 2550.



2. Create a program that computes compound interest using the for structure. Consider the following problem statement: A person invests $1000.00 in a savings account yielding 5 percent interest. Assuming that all interest is left on deposit in the account, calculate and print the amount of money in the account at the end of each year for 10 years. Use the following formula for determining these amounts: a = p (1 + r) ^n; where p is the original amount invested (i.e. principal), r is the annual interest rate, n is the number of years, a is the amount on deposit at the end of the nth year.

interest.cpp > main()

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main (){
    int principal = 1000, year;
    float rate = 0.05, amount;

    for ( year = 1; year <= 10; year++){
        amount = principal * pow(1 + rate, year);

            cout << "Year " << year << ": $" << amount << endl;
    }

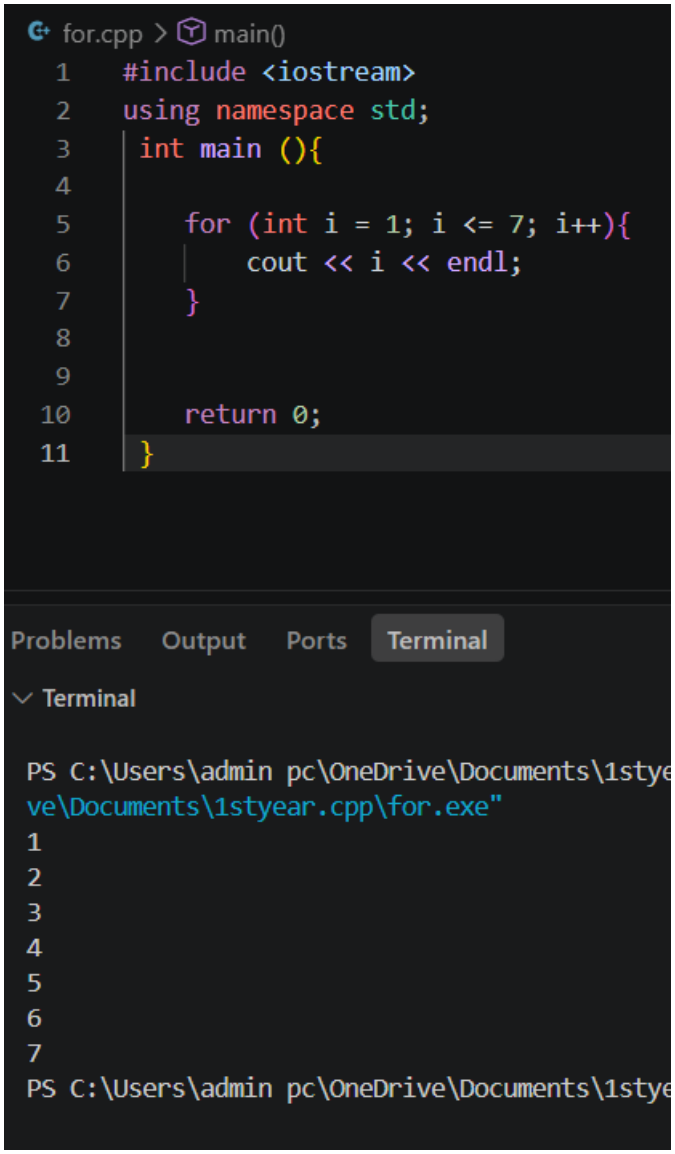
    return 0;
}
```

roblems Output Ports Terminal

Terminal

```
ve\Documents\1styear.cpp\interest.exe"
Year 1: $1050
Year 2: $1102.5
Year 3: $1157.62
Year 4: $1215.51
Year 5: $1276.28
Year 6: $1340.1
Year 7: $1407.1
Year 8: $1477.45
Year 9: $1551.33
Year 10: $1628.89
PS C:\Users\admin pc\OneDrive\Documents\1styear.cpp>
```

3. Write for statements that print the following sequences of values:
1, 2, 3, 4, 5, 6, 7



3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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